EQUIPMENT

This chapter describes some of the more common items found in an interstellar space opera setting. All prices are in Credits (Cr), the interstellar currency. Thousands of Credits are noted as KCr, and millions of Credits are noted as MCr. A Credit is a futuristic currency roughly equivalent to $3 USD in 2021CE currency.

ENCUMBRANCE

To facilitate fast play, Cepheus Deluxe avoids the careful accounting of equipment by weight. Instead, a character may carry a number of major "items" equal to 7 plus their STR DM with no penalty. This represents a Light Load. Characters may carry items beyond this, up to three times the base amount, which constitutes a Heavy Load. A heavily loaded character suffers DM-2 to all physical task throws and can only use a maximum of one move action in a combat round.

An "item" is anything of significant size: a gun, a medkit, a standard length of rope. Larger items may count as more than one item as noted in their description. Small items, such as a compass, a needle, or a candle, do not count as items, subject to the Referee's judgment.

TECH LEVELS

Cepheus Deluxe assumes an interstellar society – as befitting a space opera game. However, not all worlds in such a universe possess the same level of technological development, particularly in frontier or post-apocalyptic settings. Furthermore, Referees may want to choose between grittier, nearer-future interstellar sci-fi and no-holds-barred far-future galactic space-opera. For this purpose, Cepheus Deluxe ranks civilizations according to 17 tech levels, from 0 to 16. We present some rough historical comparisons with our own world to assist Referees and players in their understanding of tech levels. However, these comparisons should not imply that your own game must follow the same technological development.



Tech Levels

Tech Level	Notable Characteristics	Historical Comparison
0	No technology, equivalent to the Stone age. Fire, agriculture, and stone tools.	Agricultural revolution, Stonehenge
1	Roughly on a par with Bronze or Iron age technology. Writing and mathematics.	Ancient Egypt, Classical Greece, Rome, and China.
2	Late Medieval technology. Simple black-powder firearms, ocean-going vessels	Renaissance Europe, early colonial America
3	Mass production allows for product standardization, heralding the beginnings of industrial revolution and steam power.	Great Britain circa 1770
4	The transition to industrialization is complete, bringing plastics, radio, and other such inventions.	Western Europe circa 1900
5	Widespread electrification, telecommunications, and internal combustion engines.	The First and Second World Wars
6	Development of fission power and more advanced computing.	Early Cold War
7	Can reach orbit reliably and has telecommunications satellites.	We are here
8	Possible to reach other worlds in the same system, although terraforming or full colonization is not within the culture's capacity.	-
9	Development of crude gravity manipulation and controlled nuclear fusion, which makes space travel vastly safer and faster; first steps into Jump Drive technology. The advent of a wide range of cybernetics.	-
10	Development of gravitic vehicles revolutionizes transportation and warfare.	-
11	The first primitive (non-creative) artificial intelligences become possible in the form of "low autonomous" interfaces, as computers begin to model synaptic networks.	-
12	Weather control revolutionizes terraforming and agriculture.	-
13	Powered Armor appears on the battlefield in response to new weapons. "High autonomous" interfaces allow computers to become self-actuating and self-teaching. Gravitics become capable of small-scale manipulation of matter.	-
14	Fusion tools and weapons become man-portable.	-
15	The development of longevity treatments means that the human lifespan is now vastly increased.	-
16	Handheld plasma weapons and energy shields revolutionize warfare.	-

LIVING EXPENSES

Characters not embarked on a starship must pay for lodging and food. The following table provides costs of living. One column shows a monthly cost of living – assuming long-term purchase of lodging and groceries. The second shows daily living, whenever characters or tourists pay for a motel or hotel room and buy prepared food. These costs include various expenses such as taxes and transportation. Characters should pay according to the quality-of-life representative of their SOC characteristic. Upper class characters living in squalor suffer penalties, as noted below.

Living Expenses

Quality of Living	Monthly Cost	Daily Cost
Starvation (SOC 0-2)	Cr200	Cr8
Subsistence (SOC 3-5)	Cr500	Cr25
Ordinary (SOC 6-8)	Cr1000	Cr50
High (SOC 9-11)	Cr5000	Cr300
Elite (SOC 12-15)	Cr15000	Cr1000

Starvation living means living in the streets, in a tent, or outdoors and eating just barely enough to keep the character alive. Clothes are basic and always second-hand or worn out. Healthcare, entertainment, and transportation are not included in this level. A character living at starvation levels is always Fatigued.

Subsistence living is the bare minimum for civilized life – a tiny apartment, shanty, or capsule in a "coffin hotel" and simple food and clothing. Subsistence level living includes public transit, minimal entertainment, and generally inadequate healthcare.

Ordinary living is how most people live in city centers: an apartment, public transit or a small personal vehicle, reasonable food and clothing, reasonable entertainment, and modern healthcare.

High living is the good life: a fancy apartment in the city or a small house, eating at restaurants for most meals, a good personal vehicle, partying on a regular basis, fashionable clothes, and good healthcare.

The **Elite** life means living like a king: a mansion or penthouse, regular feasts, or upscale restaurants, dressing in the finest clothes, and maintaining a prestigious personal vehicle. Healthcare is the best money can buy.

A character with SOC 10+ who does not live at a High or Elite level will suffer DM-2 to all high society related social skill rolls and may be ostracized by their high-class Contacts (at the Referee's discretion).




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PERSONAL ARMOR

Personal armor reduces the damage caused by incoming attacks. Subtract the armor's Protection rating from any damage caused to the character. It is not possible to wear more than one type of armor at a time, other than energy shields, which you may use together with any other armor.

Personal Armor

Armor	Protection	Tech Level	Cost	Encumbrance	Skill Required
Civilian/Paramilitary Protection					
Leather jacket	2	1	Cr50	1	-
Heavy duster	3	6	Cr75	1	-
Bulletproof vest	4	7	Cr150	1	-
Synthsilk armor	8	12	Cr6000	0	-
Energy shield	Special	16	Cr20000	1	-
Utility Suits					
Envirosuit	3	7	Cr700	2	-
Space suit	6	7	Cr2000	2	Zero-G 0
Space suit, advanced	6	9	Cr4000	0	-
Space suit, hostile environment	10	12	Cr10000	4	Zero-G 1
Envirosuit, advanced	5	12	Cr5000	0	-
Combat Armors					
Heavy ballistic cloth	6	7	Cr250	2	-
Encased armor suit	9	8	Cr5000	2	Zero-G 1
Powered armor	18	13	Cr200000	0 (when worn)	Zero-G 2



Armor Types

Civilian and Paramilitary Protection consists of those armors that are either highly concealable, or that are acceptable to wear in most civilian settings. Characters wearing Utility Suits would never get a second look on the job where those suits are needed but wearing such a suit everywhere would raise eyebrows. Combat Armor is never discreet: whoever is wearing combat armor is signaling everyone around them that they are looking for trouble.

Bulletproof vest: lightweight ballistic armor torso protection, easy to wear under clothes.

Encased armor suit: heavy, full body rigid armor, often used by corporate SWAT and high-end mercenaries. The suit offers environmental protection as an Envirosuit (see below) and includes integrated comms and a HUD.

Energy Shield: a high-energy shield emitter worn on a belt clip, capable of stopping massive amounts of damage. The Energy Shield may be worn with other armor. An energy shield completely negates the first hit it receives in combat but is then depleted. Throw 1D every following combat round; on 4-6, the shield regenerates on your turn, and may negate another hit afterwards.

Envirosuit: a full body sealed suit with attached filters and an option to attach oxygen tanks as well. Fully protects against airborne toxins and pathogens. Reduces radiation by 20 rads. The Envirosuit counts as two Items for encumbrance purposes.

Envirosuit, Advanced: as an Envirosuit, but ultra-light. It does not count as an item for encumbrance purposes.

Heavy ballistic cloth: modern heavy-duty ballistic-cloth armor. Commonly worn by soldiers, well-equipped security, and TL7 mercenaries. Counts as two Items for encumbrance purposes and is not concealable.

Heavy duster: long overcoat made of reinforced leather or synth-leather. In some circles, this is the epitome of style.

Leather jacket: a heavy leather or synth-leather jacket.

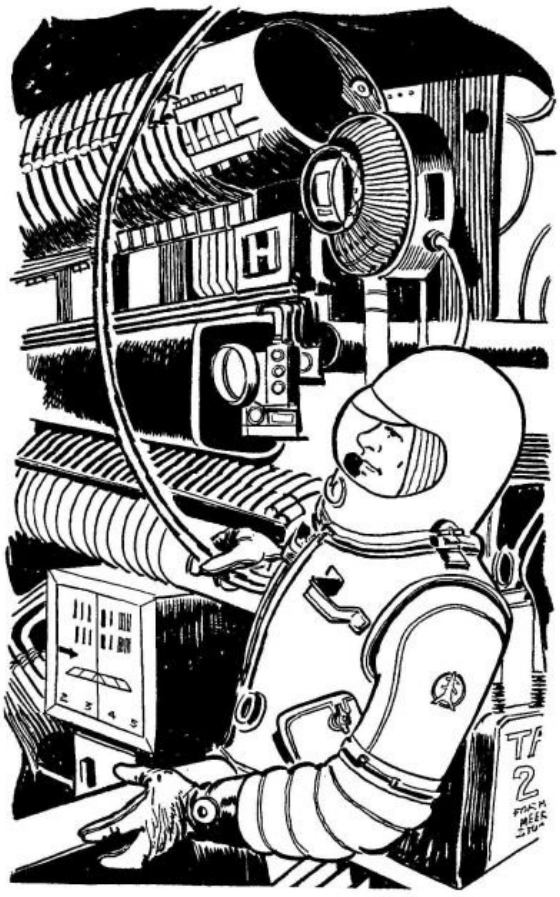
Powered Armor: servo-assisted fully enclosed combat armor. This powerful armor Grants DM+2 to STR and DEX characteristic checks and allows carrying double the user's Encumbrance. Powered armor grants DM+4 to all melee damage done by its wearer. Powered Armor protects against vacuum and hostile environments, as a space suit, with 6 hours of life support, and reduces radiation by 50 rads. Powered armor does not count as an item for encumbrance when worn.

Space Suit: a standard, futuristic space suit, lighter and cheaper than a 20th Century Terran space suit. A space suit carries life support supplies for up to 6 hours. It protects against Vacuum, Tainted, and Exotic atmospheres, as well as airborne pathogens. Offers 2D hours of protection against Corrosive atmospheres, and 1Dx10 minutes protection against Insidious atmospheres. Reduces radiation by 60 rads. Counts as 2 items for Encumbrance purposes.

Space Suit, Advanced: the final evolution of the space suit. An advanced space suit does not count as an item for encumbrance, is easy enough to use even without a Zero-G skill and provides life support for 12 hours. Offers the same environmental protection as a regular space suit.

Space Suit, Hostile Environment: a heavy-duty space suit designed for extreme conditions, such as Corrosive and Insidious atmospheres. Protects against Vacuum, Tainted, Exotic, and Corrosive atmospheres, as well as airborne pathogens and toxins, and provides 2D hours of protection from Insidious atmospheres. Carries life support supplies for 6 hours. Reduces radiation by 150 rads. Counts as 4 items for Encumbrance purposes.

Synthsilk Armor: advanced bio-engineered spider-silk armor. Synthsilk is easily concealable and does not count as an item for Encumbrance.



CYBERNETICS

Cybernetic implants, a major theme in many science fiction tales, replace or upgrade body parts with biomechanical technology. Using advanced material science and cutting-edge knowledge of the neural system, cybernetics augment the organic body, adding capabilities beyond those given to it by evolution.

CYBERNETIC GRADES AND POINTS

Cybernetics in Cepheus Deluxe come in four grades, detailed below. Each cybernetic implant cost Cybernetic Points. The implant's Grade is the primary determiner of the number of Cybernetic Points it uses. Characters may have up to 6 Cybernetic Points' worth of implants without penalty. Any more cybernetics, and the character will suffer from cyber-dissociation – being more machine than human. Cyber-dissociation applies a DM-2 penalty to all social skill throws (using SOC or any other characteristics). The only exception is when the character intends to intimidate others. In that case, there is no penalty.

Cybernetic Grades and Points

Grade	Cybernetic Point Cost	Typical Cost	Description
A	0	Up to Cr1000	Common implants used by civilians and typical corporate employees. Many of these can be implanted in out-patient procedures, depending on the world's tech level.
B	1	Up to Cr10000	Prosthetics and more specialized personal implants. Includes street-level combat implants that may or may not comply with the Law Level of a given world.
C	2	Up to Cr100000	Highly specialized implants, rare among civilian populations, including military-grade combat implants, and other sophisticated cybernetics.
R	3	Over Cr100000	The most advanced and invasive implants: the expertise alone required to install these cybernetics makes them exceedingly rare. The expense only compounds their rarity.

A-GRADE CYBERNETICS

These implants are either superficial, or extremely common among the public on most advanced worlds as to be ubiquitous.

Cyberfinger (TL9; Cr500): cybernetic finger replacement.

Glow Tattoos (TL9; Cr100): these are advanced tattoos consisting of smart bioluminescent skin grafts in complex patterns across the subject's skin. Patterns and designs are up to the subject. The subject may activate or deactivate the tattoos at will.

Internal OmniComm (TL9; Cr100): as an OmniComm (see below) but with a neural interface; this implant allows tele-video communication in the user's mind, without audible speech.

Internal OmniComp (TL9; Cr300): as an OmniComp (see below) but with a neural interface. The user can use their minds to operate the OmniComp without any gestures or speech. The details and consequences of having a computer implanted in a character's brain and connecting to planetary computing grid are left to the players and Referee.

Internal Storage Pouch (TL9; Cr800): abdominal pouch accessible via an artificial sphincter. Stows items up to 0.5 kg in weight and 0.5 liters in volume. The Storage Pouch is discreet and can only be discovered by detailed inspection or medical scans.

B-GRADE CYBERNETICS

These cybernetic implants include most prosthetics to replace limbs or organs lost to injury, or more advanced and invasive devices that greatly expand human capabilities. Cybernetic prosthetics do not enhance the character's innate characteristics—they only perfectly replace damaged parts. Some of these cybernetics are regulated by local Law Levels, while others are not.

Artificial Gill (TL9; Cr2500): allows breathing underwater. The artificial gill is powered by the body's own electrical current, and thus requires minimal maintenance. The gills can be hidden by default, though some may want them to be visible.

Cybernetic Eye (TL9; Cr5000): a baseline cybernetic eye replacement, providing perfect 20/20 vision. This cyber-eye looks indistinguishable from a normal eye. However, many opt to add cosmetic changes to the eye, or even exposed chrome to the eye housing.

Cybernetic Eye, Advanced (TL11; Cr7500): this implant operates as the baseline cybernetic eye, but also adds telescopic vision, infra-red imaging, and low-light optics.

Cybernetic Arm (TL9; Cr6000): baseline cybernetic arm replacement. The arm can be customized to look however the character likes. Human-looking cybernetic arms can pass visual inspections but are detected by medical or detailed physical inspections.

Cyberhand (TL9; Cr1500): cybernetic hand replacement. The hand can look however the character likes.

Cyberleg (TL9; Cr9000): baseline cybernetic leg replacement. Cybernetic legs are large enough to install an Internal Storage Pouch at regular prices.

Data Safe (TL9 Cr5000): this small implant stores encrypted files in the character's skull. The data safe cannot be detected by normal sensors; a deep medical scan of the character's brain is required to detect the safe. The character may or may not have access to the data safe, or the decryption key. On worlds with Law Levels 6+, the Data Safe is a regulated cybernetic.

Head Bomb (TL9; Cr3000): a tiny remote-controlled explosive device implanted in a subject's brain. The detonation will kill the subject without causing any collateral damage. Head bombs are rarely implanted voluntarily, and are widely illegal.

Internal Blade (TL9; Cr5000): a set of two 10cm retractable blades are implanted in the character's limbs. These can be wrist blades, knuckle spikes, knee spikes, or even dew claws. The character's unarmed attacks do 2D damage. The blades are constructed out of ceramic composites to avoid detection by most security scans. Only medical scans or TL13 or better sensors will detect the blades. Internal blades are considered illegal on world with Law Level 4+.

Internal Weapon (TL9; Cr5000, special): this arm-mounted implant allows the character to carry a concealed weapon internally. This can be any pistol-sized weapon limited by tech level: stealth pistol, gyrojet pistol, gauss pistol, laser pistol, blaster pistol, or stunner. The weapon's magazine capacity is half normal but can be reloaded normally via a retractable magazine holder. Internal weapons still suffer the DM-2 penalty to hit while in melee combat. The internal weapon's price is Cr5000, plus twice the weapon's cost. Internal weapons can be detected with medical scans or detailed physical inspections. Internal weapons are considered illegal on worlds with Law Level 4+.

Internal Filtration System (TL9; Cr3000): protects from airborne toxins and pathogens.

Internal Oxygen Supply (TL9; Cr4000): provides 3 hours of oxygen.

Prehensile tail (TL9; Cr9000): Provides an additional grasping/manipulating appendage. This appendage provides a DM+1 to any opposed grapple checks in melee combat but does not provide any additional actions. The tail is also strong enough to support the weight of its owner.

C-Grade Cybernetics

These implants are all very invasive, and require the amputation of two limbs, and a great deal of surgery. They are thus almost always restricted and very expensive. C-grade implants are usually reserved for elite military and espionage operatives. Only the most well-connected and wealthy civilians can acquire them.

Advanced Cyberarms (TL9; Cr20000): This implant replaces both arms, as well as reinforces the spine. Advanced cyberarms give Advantage to all arm/manipulation Athletics throws and DM+4 to unarmed combat damage. Note that this bonus applies to relevant Athletics tests, but not other skill tests. The damage bonus is cumulative with any internal blade damage bonus, and with the powered armor bonus. Advanced cyberarms count as two implants (total 4 cybernetics points), as they replace both arms.

Advanced Cyberlegs (TL9; Cr50000): this implant replaces both legs and reinforces the hips and spine. Advanced cyberlegs give Advantage to all Athletics checks involving running or leaping, subject to the same limitations as Advanced cyberarms, above. As an action, the character may throw Athletics 8+ to jump 10m horizontally in any direction or 3m upwards. Treat the character's STR as 4 points higher for purposes of calculating encumbrance limits. This implant also doubles the character's movement speed to 20m per action (up to 40m per round if the character uses both actions for movement). Advanced cyberlegs also reduce falling damage by 1D. Advanced cyberlegs count as two implants (total 4 cybernetics points), as they replace both legs.

Augmented Joints (TL9; Cr25000): an invasive series of implants replace the character's joints with advanced synthetics, high-tech lubricants and shock absorbers. Augmented joints grant inhumanly fine motor precision. This gives Advantage for non-combat throws that require fine motor manipulation and control. For example: lockpicking, sleight-of-hand, and electronics repair..

Hacking Module (TL9; Cr50000): this implant consists of an upgraded, specialized OmniComp and dedicated enhanced synaptic neural processors for computing. A discreet neural jack line feed allows the character to connect to any computer directly. The Hacking Module grants a flat DM+1 to the character's Computer skill and allows remote hacking of unprotected electronics at up to 3m. Note that at TL9+, most personal electronics cannot be remotely hacked. The Hacking module is illegal on worlds with Law Levels 4+.

Hypercosmetic Surgery (TL9; Cr75000): perfect body sculpting and implanted synthetic pheromone glands radically reconstructs the subject's body to be perfectly attractive to other members of their species. Grants Advantage to all social skill throws in situations where physical attractiveness matters.

Subdermal Armor (TL9; Cr15000): keratin and ceramic bone lacing and light shielding of vital organs provides 2 points of armor protection. The implant is subtle enough to pass visual inspection but can be spotted with sensor scans. Subdermal armor is added to the protection rating of any worn armor. Subdermal armor is legal on most worlds, but usually will be controlled on worlds with Law Level 8+.

Subdermal Armor, Heavy (TL9; Cr20000): unlike the lighter variety, heavy subdermal armor makes no attempt at being discreet. It provides 5 points of armor protection that is added to the protection rating of any worn armor. Heavy subdermal armor is detectable upon visual inspection and counts as two implants (4 Cybernetic Points). Heavy subdermal armor is illegal on worlds with Law Level 3+.

R-Grade Cybernetics

Extremely invasive, experimental, and irreversible cybernetics. These implants are all highly illegal for anyone to own, outside of restricted military circles.

Berserker Module (TL9; Cr200000): this dangerous combat augmentation keeps the user going no matter what injury they sustain. The character stays conscious and active as long as they have even one point of Lifeblood left – even when severely injured. They still die if their Lifeblood reaches zero.

Bastet Frame Upgrade (TL9; MCr1): a radical reconstruction of the entire organic body to grant superhuman agility and speed. The implant doubles the character's DEX (to a maximum of 20), and grants Advantage on all DEX-related Athletics and Stealth throws. Furthermore, the Bastet Frame increases the character's movement speed from 10m per round to 15m per round. The Bastet Frame Upgrade is incompatible with the Hercules Frame Upgrade.

Control Rig Implant (TL9; MCr1): this is a highly specialized implant designed to give elite vehicle pilots even more of an edge. The control rig allows the operator to fully interface their minds with vehicles. The control rig gives the character Advantage on Vehicle skill throws with suitably modified vehicles. This does require the operator to be in the vehicle, and "plugged in." Modyfing a vehicle to work with control rig increases its cost by 10%. If the vehicle takes damage, roll END 6+ to avoid taking 1D damage from electrical feedback.

Hercules Frame Replacement (TL9; MCr1): a radical reconstruction of the entire organic body grants superhuman endurance and might. The Hercules Frame doubles the character's STR and END (to a maximum of 20), affecting Stamina, Lifeblood, and unarmed combat damage. It also gives Advantage on all Athletics throws requiring brute strength, and Grapple checks in Melee Combat. The Hercules frame upgrade is incompatible with the Bastet Frame Upgrade.

Military-grade Neural Cyberspace Interface (TL9; MCr1): the ultimate interface between the human brain and a machine. The neural cyberspace interface functions as a Hacking Module, but also grants Advantage to all Computer and Repair throws related to electronics and computers.

Minerva Upgrade (TL13 MCr3): this implant requires the radical reconstruction of the entire organic neural system. It includes synaptic enhancers, nano-tech upgrades to dendrites, and multiple supplemental neural processor implants directly in the brain. The end result is superhuman intelligence, doubling the character's INT (to a maximum of 20).

Neural Booster (TL9; MCr2): this invasive augmentation to the character's nervous system and adrenal glands greatly boosts reaction times and suppresses the fight-or-flight response. The Booster grants an additional action per combat round for a total of three. Characters with the Neural booster implant also roll to avoid Surprise with Advantage. As well, the character never suffers the unskilled penalty (DM-3) when rolling to avoid Surprise.

Vacuum Adaptation (TL13; MCr1): the ultimate spacer implant turns the user into a voidwalker. Internal oxygen reserves, micro-seals, and metal weave skin allow an hour of operation in vacuum without a space suit. An integral grav unit allows short-range maneuvering in low and micro-gravity. The grav unit is too weak to maneuver in any gravity beyond 0.05g. This augment also provides 5 points of integral armor and reduces radiation by 10 rads. The Vacuum Adaptation implant is visible upon medical inspection or sensor scan, but is otherwise quite discreet.

Exploration and Personal Equipment

The following items are common among adventurers and explorers. This list is not complete; we invite you to add interesting equipment for their adventuring characters from other games, with Referee approval.

Exploration and Personal Equipment

Item	Tech Level	Cost
Advanced Base	8	Cr50000
Artificial Gill	9	Cr4000
Awareness Disruption Field Generator (AwFiG)	16	Cr20000
Backpack	1	Cr50
Binoculars	5	Cr10
Breather	6	Cr75
Cargo Monitor Tag	10	Cr5
Cart, Contra-Grav	13	Cr50000
Cloaking Device, Personal	16	Cr100000
Cold Weather Clothing	1	Cr200
Corrosive Gel	10	Cr250
Compass	3	Cr5
Disguise Kit	3	Cr300
Exoskeleton	8	Cr15000
Field Rations, 5 days	0	Cr125
Firestarters	8	Cr40
Grav Lamp	13	Cr100
Hammock	3	Cr25
Handcuffs	5	Cr10
Hooligan Tool	8	Cr150
LED Flashlight	7	Cr5
Lockpicks	5	Cr25
Long-Range Communicator, 500km range	7	Cr250
Mapper, Instant	9	Cr1500
Measuring Beacon	12	Cr1500
Measuring Tape, Advanced	9	Cr500
Medkit	8	Cr100
Memory Randomizer	16	Cr10000

Exploration and Personal Equipment (cont.)

Item	Tech Level	Cost
Mess Kit, Individual	4	Cr20
Multiscanner	12	Cr2000
Night Visor	8	Cr100
OmniComm	8	Cr50
OmniComp	8	Cr250
Omnitool	8	Cr150
Oxygen Tanks, 6-hour duration	6	Cr250
Panic Button	14	Cr1000
Portable Fusion Generator	12	Cr500000
Portable Generator	5	Cr1000
Pulverizer	8	Cr40
Radar Mapper	8	Cr1000
Radiation Detector	10	Cr100
Rescue Bubble	8	Cr900
Rope, 70m	1	Cr50
Sleeping Bag	5	Cr50
Sonar Window	11	Cr900
Spray Sealant	9	Cr30
Super Adhesive	8	Cr20
Super Lubricant	8	Cr15
Tech Toolkit	7	Cr300
Tent	1	Cr100
Tent, Pressure	8	Cr2000
Thruster Pack	8	Cr200
Water Canteen	1	Cr5
Water Purification Tablets	7	Cr10
Water Saw	8	Cr20000
Welder, Laser	9	Cr6000

The **Advanced Base** is pressurized, modular, and capable of housing 6 persons. It requires 12 person-hours to assemble. Life support lasts for 24 hours without portable generator to power atmoscrubbers.

An **Artificial Gill** allows unlimited underwater activity on worlds with atmospheres 4-9.

An **Awareness Disruption Field Generator (AwFiG)** emits a low-grade psionic wave that affects the perception of those around, causing them to disregard or overlook the presence of a person or thing. To the passer-by the wearer of the AwFiG is not present or is just part of the background and is unlikely to be remembered.

A **Backpack** does not count as an item for encumbrance when worn. The items in the backpack still do count towards encumbrance.

A **(Personal) Cloaking Device** permits its user to become practically invisible. The light-distorting gravitic lensing of the device affects all visible light for organic and mechanical eyes. However, the target still emits infrared heat radiation.

Cold Weather Clothing protects against inclement weather. The TL12 version costs Cr800 and does not count as an item for encumbrance.

A standard set of **Lockpicks** in a handy carrying pouch. Attempting to pick a lock without lockpicks penalizes the attempt by DM-2.

A **Contra-Grav Cart** is too slow to use as a vehicle but can carry up to 500kg of gear at a speed of 10m per combat action. With some versions, it is possible to "tether" the cart to a person, which makes the cart follow that specific person at a distance of 3m to 20m.

Corrosive Gel comes in small metallic tubes. It can be used to mark surfaces or to destroy small objects (like door locks). The gel consists of a binary chemical agent in two chambers in the tube. Once opened and applied, the agents combine within 1D minutes. The exact effect is left to the Referee. Each tube is one-use only.

An **Exoskeleton** grants its user a +6 bonus to their STR score for Athletics/STR skill checks. It also allows the character to jump up to 2m vertically, or 5m horizontally, and increases the wearer's movement speed to 15m/action. Another benefit is that the exoskeleton allows soldiers to march their normal cross-country speed while tiring at half the rate. The exokeleton's batteries are good for 48 hours of activity before requiring replacement. At higher tech levels, exoskeletons are often replaced by cybernetics.

5-days' worth of **Field Rations** each count as a single item for encumbrance. TL9+ field rations are much lighter: 25 field rations count as one item.

Firestarters are fast-burning chemicals in one-use packages. They will automatically start a sustainable fire with prepared wood and brush kindling. Throw Survive/INT 6+ to start the fire if the wood or other fuel is wet.

A **Grav Lamp** is a floating hands-free light source. It can illuminate an area between 2m and 10m in radius. The illumination radius is adjustable by means of a small lightweight "anchor" worn on the person and does not count as an item for encumbrance. The grav lamp will float near or next to the anchor, between 0.5m and 10m away, depending on the setting used.

A **Hammock** is a cloth or synthetic textile temporary sleeping surface, suspended between two fixed points. It can also be used as a shade. Supports up to 150 kg.

The **Hooligan Tool** is a 0.8-meter, 4 kg tool is designed to breach common door/hatch types. There is forked spade end (for getting doors open), a horn (for getting that initial grab into opening an iris valve or busting padlocks), and a wedge (for those pesky hatches). The tool is designed to maximize leverage. Made of special alloys and composites, the tool is non-sparking, a feature that prevents accidents in explosive atmospheres such as a ship with a liquid hydrogen leak. Complete with fiberglass shaft and hand grips made from non-conducting material.

An **LED Flashlight** is a hand-held, rubberized, sturdy light source with a variable light aperture. The 2m cone of light has a range of 50m, but it can be adjusted to illuminate a 5m radius. Smaller, more expensive (Cr20) versions can be mounted on weapons or clipped to armor. A TL7 LED Flashlight's battery will last for 24 hours. By TL11, the battery will last for a decade of continual use.

A **Mapper (Instant)** can be attached to any hand computer or used alone. It consists of a computer, display, and launcher for a data-gathering probe. The probe is an anti-grav unit that will rise 100-200 m into the air and take 360-degree image of all visible terrain. This data is then compiled into a 2-D, or 3-D map displayed by the base unit. The Mapper also instantly calculates topographic features and distances to user-selected points.

A **Measuring Tape (Advanced)** consists of a base unit and one or more movable beacon units. When activated, the device displays the exact distance between itself and the beacons. It can be thus measuring the dimensions of any object, by placing the base at one edge and the beacon at the other, or placing two beacons on either side, or measure area by placing beacons on 4 sides.

Measuring Beacon: A larger-size version of the Advanced Measuring Tape, which can be used as a homing beacon. By leaving the base at a predetermined point and carrying the beacon, the user will always know which direction and how far to go to return to the base unit. The Measuring Beacon has a 500km range.

A **Memory Randomizer** is an insidious disguise device: everyone who comes into contact with the wearer of the Randomizer perceives the wearer but cannot accurately describe them. This makes them impossible to identify by memory unless the target makes a successful INT 12+ throw. The Memory Randomizer can be placed on an object to make it so ubiquitous as to be unrecognizable.

A **Mess Kit (Individual)** includes plate, utensils, cup, storage case and disposable cleaning cloths and counts as one item.

The **Multiscanner** combines several sensor functions: it can detect radiation, electromagnetic emissions, metals, and a limited set of chemicals and biological compounds. It can also perform standard blood tests.

The **Night Visor** combines low-light and infra-red vision goggles into one handy device.

Today's smartphones will eventually evolve into an **OmniComm**, a combination of all your telecom needs in one device, with holographic augmented reality capability. The same applies to the **OmniComp** – a hand-held, light-weight computer which evolved from both the laptop and the tablet computer. Both devices are designed to integrate with whatever telecommunications networks are present on a given world. TL8 batteries for these devices last for a few days; TL9 batteries last for months between recharging.

The **Omnitool** is an auto-adjusting, auto-fitting repair tool. It will fit exactly to any kind of bolt, screw or fastening device, and also doubles as wire-cutters, small hack saw, flashlight, and a myriad of other tools. All this is packed into one item's worth of encumbrance!

A **Panic Button** uses ultrasonic waves and other sensory inputs to produce a physiological and emotional terror response. The Panic Button influences the emotional states of creatures within a 5m wide, 15m long cone, forcing them to make Morale rolls.

A **Pressure Tent** lacks an airlock and must be depressurized to enter or leave; it can house up to 2 people and requires an external oxygen tank.

The **Pulverizer** is – a combination of hammer, nail puller and pry bar. Made of high-tech materials to be lightweight but highly durable. The Pulverizer adds DM+1 to any Strength roll to force open a mechanical object such as a door or iris valve.

The **Radar Mapper** uses radar technology to scan a room or similar space, producing a 3-D map of it, to a maximum volume of 10m by 10m by 10m.

The **Radiation Detector** is a matchbox sized device that displays radiation levels in a 10m radius. It also indicates the type of radiation if that is important. The detector can also serve as an alarm, chiming loudly when radiation levels are dangerous to humans. This feature can be shut off.

A **Rescue Bubble** is a 2m-diameter plastic bubble. Piezoelectric layers translate the user's movement to electricity to recharge the bubble's battery. It contains enough life support supplies for 2 person-hours of operation. The bubble's skin can self-repair all but the most serious punctures.

A **Sleeping Bag** is a cloth or synthetic padded bag for one person, to provide insulation/ cover while sleeping. At TL 7, cold weather bags are available. Regular bags are effective to zero degrees centigrade, while cold weather bags, costing Cr150, are insulate to negative 30 degrees centigrade.

A **Sonar Window** is flat-screen device that attaches to a wall/partition. It uses sonar to display what is on the other side of the wall. Image quality depends on Tech Level. The TL11 version displays basic outlines, the TL 12 version costs Cr2000 and improves image quality and 3-D effect. At Cr7000, the TL14 version's display is virtually indistinguishable from that of an actual window.

The **Spray Sealant** can seal torn space suits and small hull breaches. The canister is good for three such applications, or one larger hull breach.

The **Super Adhesive** is instant bonding and possesses super-strength. The adhesive comes in roughly 1L canisters that are sufficient for five generous applications. The adhesive adheres to most surfaces, has a drying time of 1 minute. Pulling apart the two bonded objects requires a STR 12+ test and will ruin both objects. A special chemical is available for removing people glued to objects with Super Adhesive. It costs Cr100 per application.

The **Super Lubricant** is composed of near-frictionless space-age synthetics. Each can is good for six applications. The lubricant is a standard part of a maintenance tech's toolset. In addition to loosening stuck objects and ensuring that the internal workings of any mechanical device are working properly, the Super Lubricant can render any surface near-frictionless. Standing on a surface coated in Super Lubricant requires an Athletics/DEX 10+ throw. Walking or, heaven forbid, running on such a surface is impossible. Crawling can be done at 1m per round.

A **Thruster Pack** requires Zero-G skill level 0 and allows maneuvering in zero-G conditions at normal movement rates. Note that this is a TL8 device and does not use any anti-gravity technology. Characters and objects moving by thruster pack are still subject to the laws of motion.

Water Purification Tablets cost Cr10 for a pack of 50; each tablet can sterilize up to 1L of water from most types of microorganisms.

A **Water Saw** requires a separate power supply and a large volume of available water. Pumps create extreme water pressure (600 Mega Pascals or more) which is directed through a tiny nozzle. The stream is powerful enough to cut through concrete, stone, and even light metals. Weighs 1000kg and is usually mounted on an exoskeleton or small vehicle.

The handheld **Laser Welder** uses a focused laser beam to perform cutting and welding operations. It can cut through a reinforced bulkhead in 1D starship combat rounds (6-36 minutes). Roll Repair/DEX 10+ to reduce the time to one starship combat round (6 minutes) As a weapon, the laser welder is inaccurate and clumsy, but extremely dangerous: it does 4D damage with a range of 1/5 and has energy for 5 "shots." As a weapon, the welder also has the Fire aspect.



PHARMACEUTICALS

Below is a brief list of pharmaceuticals and drugs that are available at various tech levels.

ANTI-RADIATION

Administered before or immediately after radiation exposure, this drug absorbs 100 rads. Taking more than one dose a day causes 1D permanent END damage.

Cost: Cr1000/dose

Tech Level: TL8

CATSIGHT

This medication grants its user night vision for up to 12 hours. The character does not suffer penalties from operating and fighting in low-light conditions and only suffers DM-1 in total darkness.

Cost: Cr100/dose

Tech Level: TL9

FAST FORWARD

This drug slows metabolism by a ratio of 60 to 1 – a subjective day to the user is 2 months of real time. Note that this drug will greatly reduce life-support consumption and is often included in emergency survival kits aboard starships.

Cost: Cr200/dose

Tech Level: TL11

SLOW MOTION

This drug accelerates metabolism by a ratio of 1 to 60 – a subjective hour to the user is 60 hours of real time. Note that this drug accelerates healing accordingly. Using this drug is dangerous without Medicine 1 or better care.

Cost: Cr300/dose

Tech Level: TL11

HYPERSPACE

Hyperspace is a powerful stimulant, allowing characters to function perfectly for up to 72 hours straight without rest. Once its influence expires, the character is Fatigued (DM-2 to throws), loses 1D Stamina, and must rest for at least 12 hours to recover.

Cost: Cr50/dose

Tech Level: TL9

LONGEVITY TREATMENT

Longevity drugs prevent aging. For them to be effective, a character must take one dose each month. If the character misses even one month of Longevity Treatment, they revert to their chronological age within a week, which may be lethal in some cases.

Cost: Cr5000/dose

Tech Level: TL15

PANACEA

The Medicinal drug represents a wide variety of vaccines, antitoxins, antibiotics, and other pharmaceuticals. Panacea allows a character with at least Medicine 1 to neutralize a disease or poison on a throw of Medicine/EDU 8+. The cost of any drug is at the Referee's discretion. Advanced TL11+ medications cost ten times as much but grant Advantage on the Medicine throw.

Cost: Cr50+/dose

Tech Level: TL5



ROBOTS

Robots! The very word conjures up hopes and fears for the future. The idea of an autonomous mechanical entity is old, but its implementation became reality in the early 21st century CE on Earth (TLs7-8). At first, robots revolutionized industrial processes, but eventually even vacuum cleaners were roboticized. In these rules, contrary to various dictionary definitions, "robot" refers to a mobile, autonomous (or remote-controlled) machine capable of varied actions.

All robots and drones have a Chassis and a CPU. All robots (but not drones) have a Robot Type. It is possible to further modify robots, and we present a variety of options, but these modifications are not mandatory. Robots have a number of Build Points determined by their mass, limiting the total number of components that can be installed. For the sake of rules simplicity and clarity, like organic beings, Robots have Stamina (superficial damage capacity) and Lifeblood (internal damage capacity). They also have armor that functions like personal armor to reduce the damage of attacks.

Note that robot base speed is 5m/action. Higher speeds require the installation of Modifications. Robots start with Sensors +0, further modified by Type and specific Modifications.

ROBOT DESIGN SEQUENCE

1. Select a Chassis. This determines Mass, Stamina/ Lifeblood, Build Points, and Armor.
2. Select a CPU. Drones have no CPU; they always require a sentient operator. Smart drones and robots can operate autonomously. The CPU determines the robot's Intelligence and skill points.
3. For robots, select a Robot Type. This grants certain Modifications, two Base Skills, and could multiply the Chassis' cost.
4. In addition to Base Skills, robots with advanced CPUs have skill points to spend on skills. Use these skill points to improve Base Skills, or to purchase new ones. The maximum level in any skill depends on the CPU.
5. **Optional:** purchase modifications, by spending Build Points.

Chassis Table

Mass (kg)	Stamina/ Lifeblood	Build Points	Unarmed Damage	Base Armor	Cost (Cr)
5	1/2	1	1D-2	2	750
10 (-1 to hit)	3/6	3	1D-1	3	900
25	5/10	4	1D	3	1200
50	7/14	6	1D	4	2000
100	9/18	6	1D+1	4	4500
200	12/24	8	1D+2	5	5000
400	15/30	9	2D	5	6000
600	20/40	9	2D	8	7000
1000+	25/50	10	3D	12	10000

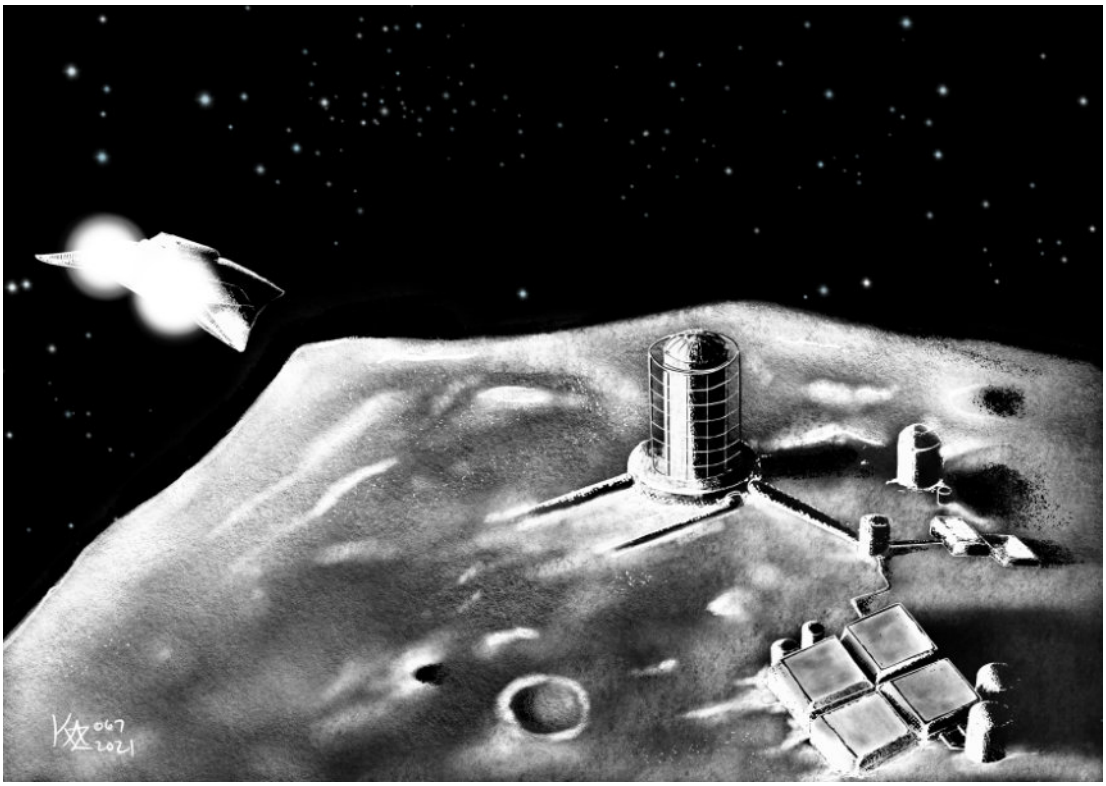
CPU Table

CPU	TL	Intelligence	Skill Points	Max Skill Level	Cost (Cr)
Drone	6	-	-	-	50
Smart Drone	8	2 (-2)	1*	1	1000
Basic Autonomous	9	4 (-1)	Base	1	5000
Advanced Autonomous	10	5 (-1)	Base+1	1	10000
Simulated Intelligence I	11	6 (0)	Base+2	2	100000
Simulated Intelligence II	13	8 (0)	Base+4	3	1000000
True AI	16	12 (+2)	Base+10	5	Priceless

* Smart Drones do not receive base skills.

Robot Types

Robot Type	Integral Modifications	Base Skills (level 1)	TL	Chassis and CPU Cost Multiplier
Combat	Heavy combat chassis Protection x2	One weapon skill, Tactics	10	X3
Humanoid	Passes as human on casual inspection; its sensors mimic human senses.	Any 2	13	X10
Janitor/Labor	-1 Sensors	-	8	X0.5
Medical	Medical Instruments cost 1 BP; DEX 8	Medicine, Science	13	X4
Protocol	-	Liaison, Admin/Steward	12	X0.75
Pursuit	Move x2	Athletics or Vehicle Skill, Tactics	9	X1
Repair	Tools cost ½ BP	Repair, Computer	10	X1.5
Scout	Sensors +2	Stealth, Recon	9	X1
Security	Sensors +1	One weapon skill, Recon	9	X1
Drone	-	Can only mount “Drone” CPUs	7	X0.5



Robot Modifications and Upgrades

Modification	Description	BP	TL	Cost (Cr)
Additional Limb	Additional arm	1	8	700
Aircraft Flight	Capable of flight; robot must be at least 50kg in mass. Can fly up to 500km/h	4	8	2500
Armor Upgrade: +1 Armor	Basic armor upgrade by 1 point of armor; max total 15	1 / +1	8	100 per point of Armor
Detachable Surveillance Module	Allows remote aerial reconnaissance	2	9	1000
Equipment	Tools or other miscellaneous equipment	1 / tool	8	Varies
Grav Module	Capable of flight, regardless of mass	1	10	10000
Holographic Camouflage	Obscures and breaks up silhouette, causing DM-2 to hit the robot.	2	12	2000
Laser Welder	Laser welder	1	8	8000
Leaper Upgrade	May jump up to double its movement range in one Action	1	9	2500
Long Range Comms	Can communicate with distant receivers, up to orbital ones.	1	8	5000
Munitions Hold	Additional ammunition for mounted weapons	1	8	100
Quad Rotors	Capable of flight; limited to 25kg or less in robot mass; can fly up to a speed of 50km/h	1	8	100
Retractable claws	Hand to hand damage 3D	2	8	150
Self-repair	1D Lifeblood repaired per round	4	13	50000
Sensor Upgrade	DM+1 to of basic sensors. Max four times.	1 / +1	8	500 per +1
Speed Upgrade: +5m/action (3 upgrades max)	Additional speed per action up to 20m/action	1 / 5m	8	100/5m
Subdermal Armor Upgrade	AR 4, humanoid robots only	2	13	1000
Enhanced Targeting Matrix	DM+1 to hit in ranged combat	1	9	5000
Tentacle	Robot gains DM+2 to Grapple throws.	2	9	1200
Weapon mount, Light	Light armament – small arm	2	8	50 + weapon
Weapon mount, Heavy	Heavy armament – heavy weapon	4	8	1000 + weapon

SAMPLE BOTS

Bot Type	TL	Cost	Type	Mass	Stamina	Lifeblood	Armor	Sensors	Speed
Cargo Bot	9	Cr6000	Labor	600kg	20	40	8	-1	5m
CPU	Basic Autonomous		**INT**	4 (-1)					
Skills	None								
Modifications	None								

Bot Type	TL	Cost	Type	Mass	Stamina	Lifeblood	Armor	Sensors	Speed
Android	15	Cr1046100	Humanoid	100kg	9	18	8	+0	10m
CPU	Simulated Intelligence I		**INT**	8 (0)					
Skills	Carousing 2, Gun Combat 1, Liaison 2, Melee Combat 1								
Modifications	Subdermal armor x1, Speed 1								

Bot Type	TL	Cost	Type	Mass	Stamina	Lifeblood	Armor	Sensors	Speed
Autodoc	13	Cr1022000	Medical	200kg	12	24	5	+3	0
CPU	Simulated Intelligence I		**INT**	8 (0)					
Skills	Medicine 3, Science 3								
Modifications	Medical instruments, DEX 8 (0), sensors +4								

Bot Type	TL	Cost	Type	Mass	Stamina	Lifeblood	Armor	Sensors	Speed
Security Bot	9	Cr6850	Security	100kg	9	18	10	+2	10m
CPU	Basic Autonomous		**INT**	4 (-1)					
Skills	Gun Combat 1, Recon 1								
Modifications	Light Weapon Mount, SMG (2D damage), Munition Hold, sensors +1, Speed 1								

Bot Type	TL	Cost	Type	Mass	Stamina	Lifeblood	Armor	Sensors	Speed
Warbot	10	Cr53850	Combat	400kg	15	30	15	+2	10m
CPU	Advanced Autonomous		**INT**	5 (-1)					
Skills	Gun Combat 1, Melee Combat 1, Tactics 1								
Modifications	2 Heavy Weapon Mounts, Light Machinegun (3D damage) and Grenade Launcher (by grenade), Munition Hold, sensors +2, Retractable Claw (3D damage)								

Bot Type	TL	Cost	Type	Mass	Stamina	Lifeblood	Armor	Sensors	Speed
Recon Drone	8	Cr3000	Scout	10kg	3	6	3	+1	10m
CPU	Smart Drone		**INT**	2 (-2)					
Skills	Recon 1								
Modifications	Sensors +1, Quad Rotors, Speed 1								

Vehicles

Science fiction stories often feature wild chases, armored warfare, or exploration of remote planets by all sorts of vehicles. A wide variety of vehicles exist across cultures and species for planetary, atmospheric, and oceanic transportation. The following is a list of a few of the more common conveyances.

Characters may purchase a vehicle at its base Tech Level or on any higher TL. Higher-tech vehicles typically have greater range and endurance, at the Referee's discretion.

Stat Block Explanation

Tech Level (TL): technological level, as noted above.

Cost: in thousands of Credits (KCr).

Agility: represents the vehicle's maneuverability. Applies as a DM to skill and Position throws in vehicle combat.

Speed: top speed in km/h; cruising speed is 75% of the listed top speed; off-road speed is equal to 50% of top speed.

Range: how far the vehicle can go at cruising speed before refueling. Given in kilometers.

Armor: the vehicle's Armor Class: Unarmored, Light, or Heavy.

Crew/Passengers: the first number shows the number of crewmembers the vehicle requires; the second, the number of passengers it may carry.

Cargo: dedicated cargo capacity in kilograms or tons.

Weapons: weapons included in the vehicle's price by default. Heavy weapons use the Heavy Weapons skill. Gunnery weapons use the Gunnery skill.

Active Countermeasures: When a vehicle equipped with Active Countermeasures is attacked by a rocket, missile, or grenade, throw 2D; on 7+, the Countermeasures intercept the munition, neutralizing the attack.

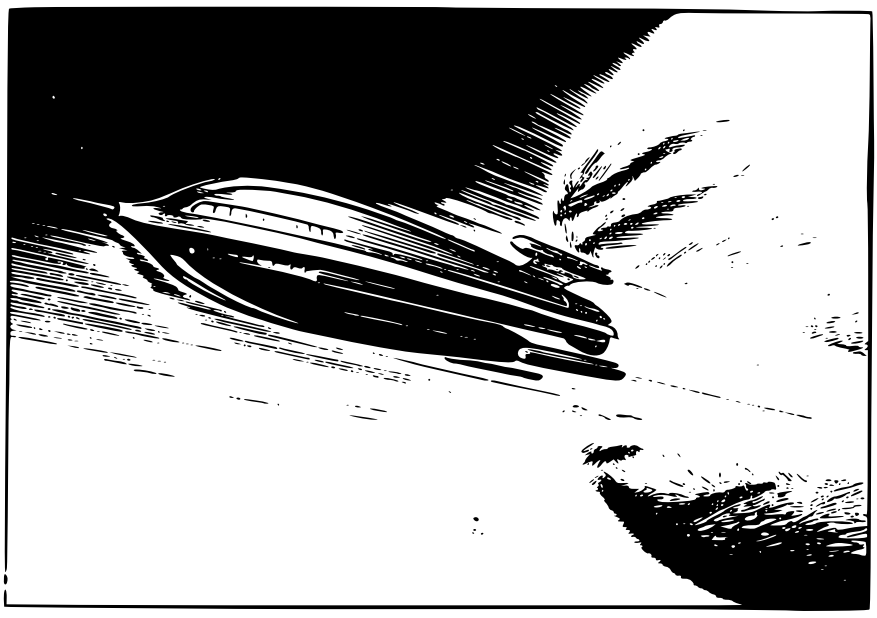
Cepheus Engine SRD/Core Conversion Note

To convert Cepheus Engine SRD/Core vehicle armor to Cepheus Deluxe, use the following table:

When using vehicles from other Cepheus Engine books that use multiple armor ratings, such as different armor for each vehicle facings, simply average the armor rating (rounding down fractions) and consult the table below.

Cepheus Engine SRD/Core Vehicle Armor Conversion

CE: SRD/Core Armor Rating	Cepheus Deluxe Armor Class
6 or lower	Unarmored
7-24	Light
25 or higher and/or Hull/Structure of 40/40 or greater	Heavy



GROUND CRAFT

4x4 Utility Vehicle			**Skill**	Driving
A heavy car (or light truck) built for ruggedness and off-road capabilities. The military version of this vehicle costs KCr84, may have Light Armor (at the Referee's discretion), and may mount two heavy weapons or one Gunnery weapon.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	40	+0	100	400
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/4	2	None	
Features: Maximum off-road speed is 75km/h.				

Armored Personnel Carrier			**Skill**	Driving
A low-tech armored vehicle used to carry infantry, that comes in either tracked or wheeled varieties. A TL8 version costs KCr200, has a top speed of 100 km/h, as well as Active Countermeasures.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
6	60	-1	70	500
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Light	3/14	1	1 GP Machinegun	
Features: None.				

Bus			**Skill**	Driving
A common mass transit bus, capable of carrying many passengers at reasonable speed and comfort.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	35	-2	100	500
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/50	8	None	
Features: None.				

Exploration Vehicle			**Skill**	Driving
A pressurized long-range heavy exploratory vehicle used for planetary surface travel and surveys.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
8	500	-1	80	500
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Light	1/14	4	None; may mount 1 heavy weapon	
Features: Maximum off-road speed is 60km/h; cramped accommodations for 15 people; sealed and pressurized; orbital-range comms array.				

Ground Car			**Skill**	Driving
The staple of low-tech urban transport. TL5-6 models are also available at an equivalent cost, but at Agility -1, speed 60 km/h, and range 200km.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	6	+0	150	400
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/4	0.2	None	
Features: None.				

Pick-Up Truck			**Skill**	Driving
An intermediate between a car and a truck. The more rugged models sometimes end up as "technicals" – improvised combat vehicles mounting a heavy weapon or even a gunnery weapon.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	10	+0	120	400
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/4	4	None; may mount 1 heavy weapon or 1 gunnery weapon	
Features: None. Technicals reduce speed to 80km/h, have no cargo space, and reduce crew/passengers to 1/2.				

Tank			**Skill**	Driving
A low-tech main battle tank. A TL8 version costs KCr2000, has an Advanced Cannon and a speed of 70 km/h, Agility -1, as well as Active Countermeasures.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
6	300	-2	50	300
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Heavy	4/0	0.5	Heavy cannon, 2 GP Machineguns	
Features: None. Off-road speed is the same as normal speed.				

Tripod			**Skill**	Driving
This ultimate alien fighting machine towers six meters above the battlefield on three thick, articulated legs, boasting massive armor and a devastating fusion cannon alongside the more mundane plasma guns. It also sports a cluster of tentacles for field work and for ripping apart enemy troops in close range.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
13	3000	+1	90	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Heavy	2/6	1	Fusion Cannon, 2 plasma guns, tentacles (Heavy Weapon for vehicle damage purposes; 4D damage to personnel)	
Features: Autopilot available; orbital-range comms; sealed and pressurized.				

Truck			**Skill**	Driving
A heavy goods vehicle. TL5-6 models are also available at an equivalent cost, but with speed 50 km/h, and range 250km.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	50	-2	100	500
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/1	30	None	
Features: None.				

Walker			**Skill**	Driving
A two-legged combat vehicle with a single pilot and heavy armor.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
13	500	+1	90	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Light	1/0	0.1	Fusion Cannon	
Features: Autopilot available; orbital-range comms sealed and pressurized.				

WATERCRAFT

Motorboat			**Skill**	Watercraft
A simple motorized boat used for fishing and short-range maritime transport.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	20	+0	60	400
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	2/10	10	None; may mount 1 heavy weapon.	
Features: None.				

Patrol Boat			**Skill**	Watercraft
A light military vessel intended for littoral (near shore) service.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	100	-1	50	500
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Light	10/20	10	2 Light Cannons; 4 GP Machineguns	
Features: None.				

Steam Ship			**Skill**	Watercraft
A low-tech maritime vessel using an external combustion engine.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
4	720	-3	20	3000
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Light	5/10	50	None; may mount up to 3 heavy weapons	
Features: 15 cabins, galley.				

Submersible			**Skill**	Watercraft
A medium-sized commercial submarine.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
6	1700	-3	40	4000
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	5/10	30	None; may mount 2 Heavy weapons	
Features: 15 cabins, galley; sealed and pressurized.				

Aircraft

Biplane			**Skill**	Aircraft
A primitive airplane with four wings.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
5	20	+0	100	400
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/1	0.5	None; may mount one heavy weapon; may carry light bombs	
Features: short-range comms.				

Bomber			**Skill**	Aircraft
A low-tech military craft used for tactical and strategic bombing.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
6	1000	-2	450	3300
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	10/0	7.5	5 GP machineguns; up to 7.5 tons of bombs	
Features: long-range comms; DM-2 attack penalty vs. propeller aircraft.				

Fighter, Propellor			**Skill**	Aircraft
A general-purpose low-tech military craft used for ground attack and interception duties.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
5	500	+0	600	750
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/0	0.5	2 GP machineguns, some might carry 1 Autocannon	
Features: long-range comms; no attack penalty vs. propeller aircraft.				

Fighter, Jet			**Skill**	Aircraft
A mid-tech, supersonic flight-capable fighter aircraft.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	10000	+1	2500	1500
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/0	1	2 autocannons; 6 air-to-air missiles	
Features: orbital-range comms; no attack penalty vs. any aircraft.				

Helicopter			**Skill**	Aircraft
A rotor aircraft capable of vertical takeoff and landing.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
6	250	-1	100	1000
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/7	1	None; may mount 2 Heavy weapons	
Features: Autopilot available; long-range comms.				

Jet Liner			**Skill**	Aircraft
A medium sized commercial passenger aircraft.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	5000	-2	1000	10000
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	2/100	20	None;	
Features: Autopilot available; long-range comms.				

Jump Jet			**Skill**	Aircraft
A commercial or paramilitary jet aircraft capable of vertical takeoff and landing.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
7	500	+1	600	5000
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	2/6	5	None; may mount one heavy weapon	
Features: Autopilot available; long-range comms.				

GRAV VEHICLES

Air Car			**Skill**	Grav Vehicle
A simple civilian gravitic car.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
10	150	+1	100	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/4	0.2	None; may mount one heavy weapon	
Features: Autopilot available; long-range comms.				

Grav Barge			**Skill**	Grav Vehicle
A commercial gravitic cargo transport				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
10	500	-1	200	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/1	50	None	
Features: Autopilot available; long-range comms.				

Grav Belt			**Skill**	Grav Vehicle
A personal, wearable means of anti-gravity mobility. Its fusion battery can last for a week.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
12	100	+2	300	1 week of use
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/0	Personal	None	
Features: Autopilot available; long-range comms.				

Grav Carrier			**Skill**	Grav Vehicle
A military gravitic flight infantry carrier.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
10	1000	+1	600	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Light	2/6	2	Light machine gun; may mount 2 more heavy weapons	
Features: Autopilot available; orbital-range comms, sealed and pressurized.				

Grav Disc			**Skill**	Grav Vehicle
A military gravitic flight infantry carrier.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
13	110	+2	400	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Light	1/0	0.05	1 plasma gun	
Features: Autopilot available; orbital-range comms, sealed and pressurized.				

Grav Gunship			**Skill**	Grav Vehicle
An advanced military gravitic attack vehicle, also capable of transporting assault troops				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
12	2000	+1	1200	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Heavy	2/12	1	Fusion Cannon, 2 plasma guns	
Features: Autopilot available; orbital-range comms, sealed and pressurized.				

Grav Liner			**Skill**	Grav Vehicle
A commercial long-range passenger grav vehicle				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
10	2000	+0	1200	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	2/100	20	None	
Features: Autopilot available; orbital-range comms.				

Grav Orb			**Skill**	Grav Vehicle
A translucent orb used for personal transportation on high-tech world. While it lacks the speed or sophisticated controls of the more advanced Grav Skiff, it offers affordable and accessible grav-flight.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
13	20	+1	100	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/1	0.2	None	
Features: Autopilot available; orbital-range comms.				

Grav Skiff			**Skill**	Grav Vehicle
A larger, faster alternative to the Grav Orb, capable of carrying more passengers and cargo. A sealed, pressurized version costs KCr375 and can carry only two passengers along with its pilot.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
13	100	+2	200	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/4	1	None	
Features: Autopilot available; orbital-range comms.				

Speeder			**Skill**	Grav Vehicle
A small supersonic grav vehicle.				
Tech Level	**Cost (Kcr)**	**Agility**	**Speed (km/h)**	**Range (km)**
10	900	+3	2000	Practically unlimited
Armor	**Crew/Passengers**	**Cargo (tons)**	**Weapons**	
Unarmored	1/1	4	None	
Features: Autopilot available; orbital-range comms; sealed and pressurized.				

Example Vehicle Agility Modifiers: note that for aircraft and grav vehicles, Agility is a function of both maneuverability and rate of climb, as both characteristics are important in Position rolls.

Example Vehicle Agility Modifiers

Vehicle Type	Agility
Slow cargo boats, submersibles, the slowest of heavy vehicles	-3
Tanks and other heavy tracked vehicles, heavy cargo aircraft	-2
Trucks, light tracked vehicles, small civilian aircraft,	-1
Small cars, maneuverable aircraft, grav cargo haulers	0
Fighter aircraft, Motorcycles, agile combat walkers, most grav vehicles	+1
Advanced fighter aircraft, military grav interceptors	+2
Grav speeders	+3



WEAPONS

Interstellar adventurers face their adversaries with blades, bullets, and lasers. Here we present a few common and iconic weapons used by starfarers.

STAT EXPLANATIONS

Tech Level (TL): technological level, as noted above.

Range: Effective/Maximum range in meters. To attack with a ranged weapon within effective range throw Gun Combat/DEX or Heavy Weapon/DEX 8+; beyond effective range, throw Gun Combat/DEX or Heavy Weapon/DEX 10+.

Damage: damage caused by the weapon, expressed as six-sided dice.

Magazine: The amount of ammunition the weapon holds. If the magazine requires additional time to reload, this is noted.

Ammo Cost: per magazine, in Credits.

Cost: The weapon's cost, in Credits.

WEAPON ASPECTS

Weapons have Aspects describing any special characteristics, as noted on their table.

Auto X: These weapons fire multiple rounds with every pull of the trigger, filling the air with a hail of fire. A weapon with the Auto aspect can make attacks in three fire modes: single, burst, and full auto, as described in the Combat chapter.

X AV Hits: These weapons are useful for attacking armored vehicles. X denotes the number of times damage is rolled on the appropriate damage table. See the Vehicle Combat chapter.

Blast X: This weapon has an explosive component or is otherwise effective against all targets in its blast radius. The number given is the blast radius in meters. It is possible to avoid or minimize blast damage, see Diving for Cover in the Combat chapter.

Bulky: A Bulky weapon has powerful recoil or is simply extremely heavy. This makes it difficult to use effectively in combat by someone of a weak physical stature. Characters with STR 8 or less suffer DM-2 when using such weapons. Bipod and Tripod accessories (see below) remove the Bulky aspect, as long as the weapon is stationary and mounted on the bipod or tripod.

Double Tap: the weapon can be fired rapidly, but at a severe loss of accuracy. It may fire 2 bullets in one attack to gain an additional die of damage.

Fire: On a successful hit, throw 6+ for the target to catch fire. A burning target takes 2D damage per combat round for 1D rounds, or until put out. See the Environmental Hazard Chapter for Fire effects.

Laser: At the Referee's discretion, atmospheric conditions, such as thick smoke, might incur a negative DM to laser fire.

Scattergun: Each attack fills the air with a hail of small projectiles, either flechettes or shot. Scatterguns cause 4D damage at Effective range or closer and attacks with scatterguns are made with DM+1. Anyone within 1.5m of the main target is also attacked by the scattergun effect, but at 2D damage. Beyond Effective range, scatterguns only inflict 2D damage and have no bonus to hit.

Stun: These weapons deal non-lethal damage and will incapacitate a living target instead of killing it. Damage is only deducted from their Stamina, after considering any armor. If the target's Stamina is reduced to 0, the target is incapacitated and unable to perform any actions for a number of rounds by which the damage exceeded his Stamina. Characters completely recover from Stun damage after one hour of rest.

Throwable: these weapons may be thrown; effective range equals the thrower's STRx4; maximum range equals STRx8.

Two-Handed: this melee weapon requires both hands to wield. However, all rifle-sized weapons must be wielded with two hands.

Zero-G: the weapon does not suffer a penalty for firing a weapon in zero-g conditions. Lasers automatically have the Zero-G Aspect.

MELEE WEAPONS

The following weapons use the Melee Combat skill. Melee weapons cannot harm vehicles except in special cases, which are left to the Referee's discretion.

Melee Weapons

Weapons	TL	Cost	Damage	Aspects
Axe	0	Cr25	2D	Throwable
Broadsword or Great-Axe	2	Cr300	4D	Bulky, Two-Handed
Cudgel	0	Cr10	2D	-
Cutlass	3	Cr100	3D	-
Dagger	0	Cr10	1D	Throwable
Machete	3	Cr50	2D	-
Neural Whip	13	Cr500	1D	Stun
Shield	0	Cr20	1D	-
Spear	0	Cr15	3D	Throwable
Staff	0	Cr10	2D	Two-Handed
Stun Prod	8	Cr300	2D	Stun
Sword	1	Cr150	3D	-
Unarmed attack	-	-	STR DM	-
Vibro-Blade	12	Cr500	4D	-

Axe (TL0): a one-handed axe made from metal or stone, anything from a stone-age tool through a medieval war axe to a modern fire axe.

Broadsword (TL2): a large two-handed sword, counts as 2 "Items" for encumbrance purposes. A two-handed Great-Axe uses the same stats.

Cudgel (TL0): a large club or mace.

Cutlass (TL3): A short, broad slashing sword with a slightly curved blade and basket-guard hilt. Traditionally used by marines and pirates.

Dagger (TL0): a straight, short knife.

Machete (TL3): A simple yet sturdy broad blade used for cutting foliage and foes alike.

Neural Whip (TL13): Consists of a stocky handle connected to a flexible strap. Each stroke by this weapon causes stun damage as above. However, it has two additional vicious effects. First, any target hit by a neural whip must throw END 10+ or fall to the ground suffering from nightmarish pain and be incapacitated for 2D combat rounds. Second, in addition to the stun damage, the neural whip causes 1D physical damage per hit.

Shield (TL0): A personal shield, with which characters can bash enemies for 1D damage. Anyone carrying a shield penalizes enemies by DM-1 when they attack. If a shield is used with the Overwatch action in combat, the attack is penalized by DM-2.

Spear (TL0): from a sharpened shaft to a metal-tipped spear, this is a basic, yet effective, melee weapon.

Staff (TL0): A long wooden shaft, sometimes with metal fittings, used with two hands.

Stun Prod (TL8): A high-tech riot weapon augmenting the baton's blunt force with an electric charge capable of stunning opponents.

Sword (TL1): one of any variety of medium-sized one-handed swords, from gladius (shortsword) to longsword and rapier.

Vibroblade (TL12): An intermediate-sized bladed weapon using hypersonic micro-vibrations of the blade to create a potent cutting force.

A note on unarmed damage: any character can inflict at least 1 point of unarmed damage (before armor) even if their STR DM is 0 or lower.

COMMON RANGED WEAPONS

The following weapons use the Gun Combat skill. Unless otherwise noted, consider them as Small Arms against vehicle armor.

Projectile Weapons

Weapon	TL	Cost	Range	Damage	Magazine	Ammo Cost	Aspects
Gyrojet Rifle	9	Cr900	100/400	3D	20	Cr15	Auto 2, Zero-G
Gyrojet Pistol	9	Cr400	10/20	2D	8	Cr10	Zero-G, Double Tap
Pulse Rifle	9	Cr1000	200/400	3D	40	Cr30	Auto 3
Assault Rifle	7	Cr300	50/200	3D	30	Cr15	Auto 2
Autopistol	5	Cr150	10/50	2D	16	Cr7	Double Tap
Stealth Pistol	8	Cr500	5/20	2D	6	Cr20	Double Tap
Bow	0	Cr60	300/600	2D	1	Cr10/20 arrows	-
Carbine	5	Cr250	50/200	3D	10	Cr12	Double Tap
Crossbow	2	Cr75	300/600	3D	1	Cr10/20 arrows	-
Gauss Pistol	12	Cr1000	20/100	3D	30	Cr30	Auto 2
Gauss Rifle	12	Cr1500	300/600	4D	100	Cr40	Auto 4
Light Machinegun	7	Cr1200	100/300	3D	100	Cr125	Auto 4, Bulky
Revolver	4	Cr150	10/50	2D	6	Cr5	-
Rifle	5	Cr500	200/400	3D	10	Cr15	-
Shotgun	4	Cr200	20/40	4D	6	Cr10	Scattergun
Submachinegun	5	Cr500	30/100	2D	30	Cr20	Auto 3
Tangler	12	Cr750	30/90	-	1	Cr100	-

Assault Rifle (TL7): Standard military arm on mid-tech worlds. An automatic rifle that fires an intermediate caliber. At the Referee's discretion, its ammunition, or even magazines, may be interchangeable with those of a Carbine.

Autopistol (TL5): Semi-automatic handgun fed from a magazine.

Bows and Crossbows (TL0 and 2, respectively): muscle-powered ranged weapons. Ammunition for a Bow is held in a quiver of 20 arrows, which counts as one encumbrance item. A Bow may fire once a round; a Crossbow requires 2 actions to reload; a quiver holds 20 crossbow quarrels and counts as one encumbrance item.

Carbine (TL5): military longarm that uses a smaller caliber than a full-scale rifle. Carbines are lighter and shorter than rifles, and are often employed as survival weapons.

Gauss Pistol and Rifle (TL12): These weapons accelerate small metal spikes to supersonic velocities. Though the principles behind their function have been understood since the early industrial age, Gauss weapons are typically beyond the manufacturing capabilities of frontier worlds. They do appear, from time to time, in the hands of well-equipped mercenary forces.

Gyrojet (TL8): Designed for use in zero-G environment, the gyroget rifle fires self-propelled rocket rounds. It has minimal recoil and works perfectly in hard vacuum. The Gyrojet Pistol uses the same rounds as its larger cousin, but is used as a sidearm.

Light Machine Gun (TL7): Standard mid-tech squad-level support automatic weapon, typically belt-fed. Considered as two items for encumbrance purposes.

Pulse Rifle (TL9): An advanced rifle firing electrically ignited, caseless, high explosive, armor-piercing rounds. Provides a high rate of fire and a large magazine capacity.

Revolver (TL4): Simple semi-automatic pistol using a manually loaded rotating drum to feed the bullets. A reliable weapon.

Rifle (TL5): a standard military firearm on low-tech worlds that uses a large caliber projectile. At higher TLs, up to and including TL9, such weapons often serve as dedicated sniper rifles.

Shotgun (TL4): Scattergun firing a "shot" palette capable of hitting multiple targets at once. These particulate stats represent a pump-action weapon with an internal magazine.

Stealth Pistol (TL8): This is an especially light, discreet, and concealable weapon made from advanced polymers. A concealed body pistol can also be disguised as a number of more innocuous items, like comms, digital recorders, and so on.

Submachinegun (TL5): Light automatic weapon that fires pistol rounds. Submachinegun bullets are interchangeable with those of the autopistol; at the Referee's discretion, so are the magazines.

Tangler (TL12): A stubby, tube-like weapon using a gravitic launch system to deliver a capsule to its target. On a hit, the capsule erupts into a mass of adhesive tentacle-like strands, entangling the target creature and any other creature within a 5m radius. Any creature caught in the strands must roll STR 12+ to break out. The Tangler Gun creates enough strands to entangle a creature up to 500kg in weight; larger creatures are unaffected. On a failed roll, the target is immobilized and cannot act; they may attempt to free themselves each minute (10 combat rounds). Other characters can also cut entangled targets free using edged weapons. The Tangler comes with solvent capsules – which the weapon itself can launch as projectiles – which dissolve the tangling strands holding captives. Tangler strands hit with a solvent capsule will dissolve in one minute.

Energy Weapons

Weapon	TL	Cost	Range	Damage	Magazine	Ammo Cost	Aspects
Blaster Pistol	13	Cr5000	50/200	4D	20	Cr100	Laser, Double Tap
Blaster Rifle	13	Cr10000	200/400	5D	50	Cr500	Auto 3, Laser
Laser Pistol	10	Cr1500	50/200	3D	10	Cr500	Double Tap, Laser
Laser Rifle	9	Cr3500	300/500	5D	30	Cr1500	Double Tap, Laser
Plasma Rifle	16	Cr15000	150/300	6D	10	Cr500	Fire
Stunner	10	Cr600	10/30	3D	10	Cr10	Stun, Zero-G

Blasters (TL13): are advanced laser weapons with a compact, internal battery, which is also removable and rechargeable.

Laser Pistol (TL10): Handgun-sized directed energy weapon attached to a belt-mounted power pack. The pistol and its power pack are considered one item together. The powerpack is rechargeable from any major power source, such as a ship's powerplant.

Laser Rifle (TL9): Requires a rechargeable heavy backpack power unit, which counts as a separate item. It is rechargeable from any major power source, such as a ship's power plant.

Note that **laser weapons** suffer DM-1 to hit when used in environments with a great deal of particulates in the air, like sand, or ash. In these conditions, their ranges are reduced by half as well.

Plasma Rifle (TL16): The ultimate small arm, using miniaturized plasma gun technology to propel magnetically shaped pockets of super-hot plasma at its targets.

Stunner (TL10): Firing a precise electromagnetic pulse, this non-lethal weapon disrupts the target's neural processes, leading to stun and potentially unconsciousness.



Common Small Arm Accessories

The following accessories are commonly available for ranged weapons.

You may attach a **Bayonet** (TL2), identical to a dagger in its game statistics on its own, to any rifle-sized weapon. A rifle with a bayonet serves as a melee weapon equivalent to a Spear (3D damage).

Laser Sight (TL7): the bonus for Aimed Shots increases to DM+2, but only within Effective range. The DM+3 limit on aimed shot bonuses to hit is still in effect. Cr100.

Scope (TL6): the weapon avoids the penalty for firing beyond effective range, as long as the character used the Aimed Shot action before firing. Maximum range still applies. Cr200.

Silencer (TL6): May be attached to any firearm. Silencers work for single shots, and short bursts, but never Full Auto attacks. Silencers provide a DM-4 to hear the gunshot sound. Cr250.

Smart Weapon (TL10): connects to a HUD or an implanted OmniComp. Grants DM+1 to all attacks. The Smart Weapon bonus is not cumulative with laser sights. This means that after one Combat Action aiming, the character can fire at DM+3. The DM+3 limit on aiming bonuses still applies. Cr1000.

All Heavy weapons can be mounted on a **Tripod** (TL4), which negates their Bulky aspect. The weapon cannot be moved while mounted on a tripod. Tripods take 10 combat rounds (one minute) to set up. Cr100.



COMMON HEAVY WEAPONS

The following weapons use the Heavy Weapons skill. All heavy weapons are considered 3 items each for encumbrance purposes. Ammunition belts, magazines, and individual rockets are each considered one item for encumbrance purposes.

Heavy Weapons

Weapon	TL	Cost	Range	Damage	Magazine	Ammo Cost	Aspects
Disposable Anti-Vehicle Rocket	7	Cr200	50/200	5D	1	-	Blast 1, Bulky, 2 AV hits
Flamethrower	6	Cr1000	5/20	3D	5	Cr25	Bulky, Fire
Fusion Gun	14	Cr100000	300/600	12D	-	Cr65000	Bulky, Fire, 4 AV hits
GP Machinegun	5	Cr1500	200/400	3D	100	Cr150	Auto 4, Bulky
Grav Launcher	13	Cr20000	500/1000	10D	1	Cr100	Blast 5, Bulky, Fire
Grenade Launcher	7	Cr400	100/500	By grenade	6	0	Bulky
Guided Anti-Tank Missile	7	Cr4000	200/600	5D	1	Cr200	Blast 1, 2 AV hits
Light Assault Gun	8	Cr600	100/300	5D	5	Cr30	Bulky, 2 AV hits
Plasma Gun	12	Cr20000	200/500	10D	40	Cr2500	Bulky, Fire, 3 AV hits
Rocket Launcher	6	Cr2000	50/200	5D	1	Cr150	Blast 6, Bulky, 1 AV hit
Support Laser	9	Cr5000	100/400	6D	100	Cr2000	Bulky, 2 AV hits
Under-barrel Grenade Launcher	7	Cr1000	50/200	By grenade	1	-	-

Disposable Anti-Vehicle Rocket (TL7): A cheap, one-use, portable rocket launcher. Functions as an armor-piercing rocket, with 2 AV hits against vehicles.

Flamethrower (TL6): Spews a stream of fire 3m wide at its far end. All targets within this cone suffer damage and catch fire. The tank for the flamethrower holds enough fuel for five full attacks and counts as two items for encumbrance.

Fusion Gun (TL14): The advanced development of the plasma gun. This weapon fires plasma bolts undergoing microfusion for massive damage. It uses a back-mounted micro-fusion reactor and thus, for game purposes, has unlimited ammunition. The ammo cost in this case represents purchasing such a backpack. Note that the backpack and gun constitute 4 items in total for encumbrance.

GP Machinegun (TL6): A “General Purpose" belt-fed medium machinegun. Often carried on vehicles as a heavy weapon but is also man portable.

Grav Launcher (TL13): Advanced micro-gravitic technology allows this weapon to guide a floating plasma bomb to its target without having to account for ballistic trajectories or inertia. Once launched, the bomb travels 50m per combat round. Its operator can move it in any direction and can also stop it and have it hover in mid-air. Detonation is on command. The bomb's internal battery allows for 10 rounds of gravitic flight.

Grenade Launcher (TL7): Grenade launchers fire grenades over long distances. Grenades for a grenade launcher are not interchangeable with handheld grenades.

Grenade Launcher, Under-Barrel (TL7): this weapon greatly increases the firepower of infantry. The under-barrel grenade launcher can be mounted under most two-handed firearms at TL7 or higher. It is not Bulky and can fire a single grenade before reloading. At TL10+, the weapon has a 3-shot magazine.

Guided Anti-Tank Missile (TL7): this is a portable guided missile system, like the AT-3 Sagger, or present-day Javelin or Eryx. These weapons must be fired from a fixed stationary position. However, the gunner need not be adjacent to the missile’s firing position.

Light Assault Gun (TL8): Heavy anti-vehicle rifle firing large armor-piercing shells.

Plasma Gun (TL12): High-energy anti-vehicular man-portable weapon firing magnetically contained plasma bolts. It requires a back-mounted, rechargeable power pack counting as two Items for encumbrance purposes.

Rocket Launcher (TL6): Man-portable rocket launcher utilizing unguided munitions. Rocket launchers are unsafe to use in enclosed spaces. Rocket payloads are considered high explosive and are not particularly effective against armored vehicles. Armor piercing rockets cost Cr500 per rocket: trade Blast 6 for 2 AV hits, Blast 1. An anti-aircraft homing missile (2 AV hits) costs Cr10000 and does not suffer a penalty when firing at fast-moving air vehicles.

Support Laser (TL8): Heavy directed-energy man-portable weapon for fire support and light anti-vehicular duties. Its back-mounted power pack counts as two Items for encumbrance purposes.

Grenades and Explosives

Throw Athletics/DEX 8+ to accurately throw a hand grenade within effective range, or Heavy Weapons/ DEX 8+ to accurately launch a grenade from a launcher. The Effective range for thrown grenades is equal to the character's STRx4. Maximum range for thrown grenades is STRx8.

On a failed grenade attack, thrown grenades will deviate 2D + (the margin of failure) meters from its intended target, double that for grenades fired from a launcher. Beyond Effective range a thrown grenade automatically deviates 2D – (Effect) meters, regardless of the result of the attack throw.

Throw Demolitions/EDU 8+ to properly set up an explosive charge for simple demolition, or Demolitions/EDU 10+ or greater for more complex jobs, such as setting a directional shaped charge explosive.

Grenades and Explosives

Grenade or Explosive	TL	Cost	Damage	Aspects
Dynamite	4	Cr75	3D	Blast 1D
EMP	8	Cr50	4D	Blast 5, Stun (Machines), Heavy: 2 AV hits
Flashbang	7	Cr30	3D	Blast 5
Fragmentation	5	Cr30	3D	Blast 10 (Stun)
HEAP (launcher only)	7	Cr50	5D	Blast 1, Heavy: 1 AV hit.
Incendiary	6	Cr30	3D	Blast 3, Fire
Molotov Cocktail	4	Cr5	2D	Blast 2, Fire
Plasma	13	Cr250	5D	Blast 5, Fire, Heavy: 2 AV hits
Plastique	6	Cr200	6D	Blast 10
Smoke	6	Cr15	-	Blast 10 (smoke)

Dynamite (TL4): Low-tech explosive made of nitroglycerin, sorbents (such as powdered shells or clay) and stabilizers. Used for demolition but can also be thrown with a lit fuse as a "grenade".

EMP Grenade (TL8): Electro-Magnetic Pulse grenade emitting a powerful magnetic pulse capable of disrupting electronics but harmless to organic beings. It causes full damage to robots. Characters with implanted cybernetics are vulnerable to EMP but only take half stun damage (rounded up) from it. Non-cybered organic beings are immune to this stun damage. This grenade counts as a heavy weapon (2 AV hits) for the purposes of damaging TL7+ vehicles.

Flashbang Grenade (TL7): Non-lethal stun grenade emitting a powerful flash and deafening noise upon detonation.

Fragmentation Grenade (TL5): Basic anti-personnel hand grenade exploding into a hail of shrapnel.

HEAP Grenade (TL7): High Explosive Armor-Piercing anti-tank grenade using a focused blast to penetrate armor. This grenade can only be fired from a launcher and counts as a heavy weapon (1 AV hit) for purposes of damaging vehicles.

Molotov Cocktail (TL4): Home-made incendiary bomb commonly made from a glass container filled with highly flammable liquid and lit with a fuse.

Incendiary Grenade (TL6): Military-grade fire-bomb grenade.

Plasma Grenade (TL13): A plasma explosive device in a grenade-sized package. This grenade counts as a heavy weapon for damaging vehicles (2 AV hits).

Plastique (TL6): This generic, multi-purpose plastic explosive is a favorite of military units, terrorists, demolition teams and adventurers across known space.

Smoke Grenade (TL6): Deploys a cloud of thick smoke upon impact. Apply DM-1 to all attacks passing through its Blast radius. They also apply a further DM-1 to laser weapons. The smoke stays in the air for 1D combat rounds.

GUNNERY WEAPONS

The following weapons require a vehicular or fixed tripod mount. They use the Gunnery skill. Consider them as Gunnery Weapons when fired against vehicle armor.

AV Hits represents the number of rolls on the Surface Damage or Internal Damage tables if the weapon penetrates the target. If the weapon causes Critical Damage, roll only once on the Critical Damage table, regardless of how many AV Hits the weapon usually causes.

Direct hits on personnel with Gunnery Weapons are automatically lethal, unless the target is wearing TL13 powered armor, in which case, they take the Personal Damage listed here, normally. If the character is in the blast radius of a Gunnery Weapon, apply the Personal Damage normally, and see the rules on Diving for Cover in the Personal Combat chapter to maximize chances of survivability.

Gunnery Weapons

Weapon	TL	Cost (KCr)	Range	AV Hits	Personal Damage	Aspects
Air-to-Air Missile	7	KCr50/each	2000/4000	3	4D	No penalty vs. aircraft
Anti-Tank Missile System	7	KCr15	500/2000	4	6D	Blast 3
Autocannon	6	KCr50	750/1500	1	4D	Auto 2, Blast 1
Cannon, Light	5	KCr30	500/1000	1	4D	Blast 2
Cannon, Heavy	6	KCr60	750/1500	2	5D	Blast 5
Cannon, Advanced	8	KCr120	750/1500	3	5D	Blast 5
Fusion Gun	13	KCr2500	3000/6000	6	10D	Blast 8, Fire
Laser Cannon	8	KCr150	1500/3000	3	6D	Laser
Laser Cannon, Advanced	9	KCr100	1500/3000	4	6D	Laser
Plasma Cannon	11	KCr750	750/1500	5	10D	Blast 5, Fire
Railgun	10	KCr75	2000/4000	3	8D	Blast 3

Air-to-Air Missile (TL7): this is a radar-guided or heat-seeking missile designed for use in air-to-air combat. TL8+ versions are self-guided suicide drones. These missiles can represent large surface-to-air missile installations as well.

Anti-Tank Missile System (TL7): A larger, vehicle or fixed-mount version of the guided missile heavy weapon, like the modern TOW or Hellfire missile. These missiles are very effective against slower armored vehicles.

Autocannon (TL 6): a rapid-fire cannon in the 20mm-30mm caliber range that is usually found mounted on aircraft and light vehicles. This weapon usually has enough ammunition for 10-20 short bursts.

Cannon, Light (TL5): a vehicle-mounted or towed gun in the 40-70mm caliber range, like the guns mounted on early Second World War tanks. They usually have between 40 and 100 shells in their magazine.

Cannon, Heavy (TL6): this large vehicle-mounted cannon is the main weapon of modern tank designs. This also represents field guns and light indirect artillery.

Cannon, Advanced (TL8): the final evolution of chemically-ignited direct fire heavy guns. Each projectile is an advanced charge that can be set to detonate in the air, upon contact, or designed to self-forge into an armor-piercing penetrator.

Fusion Gun (TL13): a large and bulky high-energy weapon that fires magnetically shaped bolts of plasma undergoing nuclear fusion. This weapon is powered by its own fusion reactor and thus has effectively unlimited ammunition.

Laser Cannon (TL8): the first battlefield energy weapon. This laser is more costly than an advanced cannon but has twice the effective range. Unfortunately for its operators, laser cannons suffer a DM-1 to hit when there is a large amount of particulates in the air.

Laser Cannon, Advanced (TL9): a fully developed, and thus cheaper, battlefield vehicle laser. It is still affected by particulates like all lasers.

Plasma Cannon (TL 11): The earliest high-energy weapon on the battlefield. The plasma cannon fires magnetically shaped plasma bolts that release enormous amounts of heat on the target. They are not long-ranged vehicle weapons.

Railgun (TL 10): an answer to newly developed laser cannon technology, the railgun fires a slug of nickel-alloy at hypersonic velocities. The weapon is cheap and has a very long range.